1. Preliminaries: social ranking functions and the Majority Judgment

Social choice theory, first introduced by De Condorcet (1785) and further developed by Arrow (1951), provides a rigorous framework for aggregating individual preferences, judgments, and votes into collective decisions. It can be viewed as an MCDM framework, where voters act as evaluation criteria, assessing feasible alternatives to determine the most preferred candidate. A social ranking method gathers individual opinions on each candidate and systematically aggregates them to identify the best choice or potential ties. Let us start by introducing the basic ingredients needed to construct a social ranking function. We define a finite number of competitors or candidates as $C = \{C_1, C_2, \ldots, C_m\}$, a finite number of judges or voters $J = \{J_1, J_2, \ldots, J_n\}$, and a common language of grades $\Lambda = \{\alpha, \beta, \gamma, \ldots\}$ that is a totally ordered set. Considering those inputs we can define a voting profile as an $m \times n$ matrix Φ of the grades $\Phi(i, j) = \alpha_{ij} \in \Lambda$ that all judges $J_j \in J$ assign to each of the competitors $C_i \in C$.

A *method of ranking*, $\succeq_J$, where $C_i \succeq_J C_k$ indicates that jury J prefers C_i to C_k, is defined as a complete binary relation constructed starting from May (1952) axioms. A *social-ranking function* is a method of ranking that needs to satisfy the following properties:

1. *Anonymity.* Permuting the names of voters does not change the outcome (Pareto optimality).
2. *Neutrality.* Permuting the names of candidates does not change the outcome.
3. *Transitivity.* If $C_i \succeq_J C_k$ and $C_k \succeq_J C_l$ then $C_i \succeq_J C_l$. That is, Condorcet's

paradox[1] cannot occur.

4. *Independence of irrelevant alternatives.* If $C_i \succeq_J C_k$ then whatever candidates are dropped or adjoined $C_i \succeq_J C_k$. That is, Arrow's paradox[2] cannot occur.

1.1. Majority ranking

Among the social ranking methods, this work focuses on the theory of Balinski and Laraki (2010), i.e., the majority judgment, that we outline below. Majority ranking is a multi-step process that starts with candidates being graded by a jury based on evaluation criteria. The vote/profile matrix, Φ, where judges' preferences are stated in a universal language Λ serves as a basis for defining a method of grading. This must be an aggregation function, F, summarizing all grades assigned by judges to competitors and using the same language: $F : \Lambda^{m\times n} \to \Lambda^m$. Therefore, $F(\Phi)$ is a vector whose i-th component, F_i is the aggregated or final grade assigned to competitor $C_i \in C$ by F. To be a *social-grading aggregation function*, F needs to satisfy the following properties:

- *Anonymity*: Permuting the names of judges (voters) should not affect the final grade, i.e., $\forall i \in \{1 \ldots, m\}$, $F_i(\ldots, \alpha, \ldots, \beta, \ldots) = F_i(\ldots, \beta, \ldots, \alpha, \ldots)$.

- *Unanimity*: If all judges assign the same grade to a competitor, C_i, then the aggregated grade must be the same, i.e., $F_i(\alpha, \alpha, \ldots, \alpha) = \alpha$.

[1] If this condition is violated, Condorcet's paradox occurs (De Condorcet, 1785). Suppose we have three candidates, C_1, C_2, and C_3, and three voters, J_1, J_2, and J_3, who have the following preference: $C_1 \succ_{\{J_1\}} C_2 \succ_{\{J_1\}} C_3$; $C_2 \succ_{\{J_2\}} C_3 \succ_{\{J_2\}} C_1$; $C_3 \succ_{\{J_3\}} C_1 \succ_{\{J_3\}} C_2$. In this case, the transitivity property fails, and there are cyclical societal preferences $C_1 \succeq_{\{J_1,J_2,J_3\}} C_2 \succeq_{\{J_1,J_2,J_3\}} C_3 \succeq_{\{J_1,J_2,J_3\}} C_1$. Even if the preferences of individual voters are transitive, collective preferences can be non-transitive. There is no clear winner even though each pair has a majority preference.

[2] This propriety avoids Arrow's paradox (Arrow and Raynaud, 1986), ensuring that the relative ranking of two alternatives should not change when other irrelevant alternatives are introduced or removed.

- *Monotonicity*: If every judge increases C_i's grade (or leaves it unchanged), the final aggregated grade should not decrease. The final grade should increase if all individual grades increase.

$$\begin{aligned} \alpha_j \preceq \beta_j \ \forall j \in \{1, \dots, n\} \quad &\Rightarrow F_i(\alpha_1, \dots, \alpha_n) \preceq F_i(\beta_1, \dots, \beta_n) \\ \alpha_j \prec \beta_j \ \forall j \in \{1, \dots, n\} \quad &\Rightarrow F_i(\alpha_1, \dots, \alpha_n) \prec F_i(\beta_1, \dots, \beta_n) \end{aligned}$$

By assigning a final grade to each competitor, F determines the order-of-finish of all competitors. The social grading function used by MJ is the *majority grade*, the unique middlemost aggregation function respecting consensus, i.e.,

$$F_i^{MJ} = \begin{cases} r_{(n+1)/2} & \text{if } n \text{ is odd,} \\ r_{(n+2)/2} & \text{if } n \text{ is even.} \end{cases}$$

Here, r_a denotes the a-th element of the ordered vector $r = (r_1, \dots, r_n)$, which represents a permutation of the candidate C_i's voting profile, $(\alpha_1, \dots, \alpha_n)$, arranged in non-increasing order: $r_1 \succeq r_2 \succeq \dots \succeq r_n$. Therefore, a competitor's majority grade is the grade that obtains an absolute majority of the voters against any lower grade and an absolute majority or a tie against any higher grade. The MJ grading principle acknowledges consensus: if a jury is more aligned in their assessment of one candidate compared to another, the stronger agreement should be honored by awarding a final grade that is at least equal to the other option's grade.

When aiming to rank candidate solutions, relying solely on the majority grade may be insufficient. While a candidate with a higher majority grade is naturally placed higher in the hierarchy due to the implied order of grades, there may be situations where different candidates receive the same grade. Ties between candidates are solved using the majority value, i.e., an ordered list of grades received by the candidate from the set of judges. Suppose we want to rank two candidates, C_1 and C_2, whose ordered list of grades are

$$r^{C_1} = (r_1 \succeq r_2 \succeq \cdots \succeq r_n)\,; \quad r^{C_2} = (r'_1 \succeq r'_2 \succeq \cdots \succeq r'_n)\,.$$

if their majority grades are the same, we can break the tie by removing the central entry from both vectors, re-evaluating the majority grades of the candidates, and comparing them. This procedure can be repeated until we find a mismatch in the majority grades obtained from the jury subsets. The only ties left will be those cases where $r^{C_1} \equiv r^{C_2}$ so that the two candidates are indistinguishable under the anonymity assumption. The majority value, r_v, of a candidate is the list of grades ordered to make iterative comparison straightforward:

$$r_v = \begin{cases} (r_{\tilde{m}}, r_{\tilde{m}+1}, r_{\tilde{m}-1}, r_{\tilde{m}+2}, r_{\tilde{m}-2}, \dots r_n, r_1) & \tilde{m} = \frac{n+1}{2} \quad \text{if } n \text{ is odd,} \\ (r_{\tilde{m}}, r_{\tilde{m}-1}, r_{\tilde{m}+1}, r_{\tilde{m}-2}, r_{\tilde{m}+2}, \dots, r_n, r_1) & \tilde{m} = \frac{n+2}{2} \quad \text{if } n \text{ is even.} \end{cases}$$

Thus, a competitor's majority value is the sequence that begins at the middle, $r_{\tilde{m}}$, and fans out alternately from the center starting from below (above) when there is an even (odd) number of voters.

After introducing the majority value of different candidates we are able to rank different solutions by defining the *majority ranking* criterion, $\succ_{MJ}$. Given two competitors, C_1 and C_2, with majority values $r_v^{C_1}$ and $r_v^{C_2}$ then we say that $C_1 \succ_{MJ} C_2$, when $r_v^{C_1} \succ_{lexi} r_v^{C_2}$, where $\succ_{lexi}$ means lexicographically greater. Lexicographic order is a generalization of the alphabetical order of the dictionaries to sequences of ordered symbols or elements of a totally ordered set. Therefore, the majority ranking is defined by the lexicographic order among the sequences of grades that are the majority values.

In Figure 1 we depict the steps leading to the identification of majority ranking for a simple case where three judges grade four candidates using a voting system

$\Lambda = \{1 \preceq 2 \preceq 3 \preceq 4\}$. These go from the initial grading of candidate solutions through the definition of majority values to the final determination of ranking.

Among the infinite ranking methods, the MJ stands out for its resistance to manipulations. The MJ permits voters to better express their opinions and always gives a transitive ranking of candidates, thus avoiding the Condorcet paradox. Moreover, the order between two candidates is not dependent on the grades received by other candidates, thus eliminating the Arrow paradox. It best combats voters' strategic manipulation, thereby inciting honest opinions. Furthermore, a candidate whose grades dominate another wins, preventing the domination paradox. Despite strategic voting and manipulation not being straightforwardly applicable to anomaly-based portfolio construction, these properties imply that MJ is robust against the presence of outliers (extreme votes) or systematic, "non-random" vote distortions (bias), rather common peculiarities of time-varying financial data.

	J_1	J_2	J_3
C_1	1	3	2
C_2	4	2	3
C_3	2	1	1
C_4	3	3	4

$\rightarrow$

	r_1	r_2	r_3
C_1	3	2	1
C_2	4	3	2
C_3	2	1	1
C_4	4	3	3

$\rightarrow$

	r_v
C_1	[2,1,3]
C_2	[3,2,4]
C_3	[1,1,2]
C_4	[3,3,4]

$\rightarrow$

	r_v	**rank**
C_4	[3,3,4]	1
C_2	[3,2,4]	2
C_1	[2,1,3]	3
C_3	[1,1,2]	4

Figure 1: Transformation steps leading to majority ranking definition. The original grade profile where three judges J_i grade four candidates C_j (leftmost) is transformed into a sorted equivalent profile (center-left) through anonymity. The majority grade is identified by the central column (light gray). Subsequently, the majority value sequences are computed (center-right). The lexicographic order of the candidate majority values determines the majority ranking (rightmost).

References

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